No. of Printed Pages : 6 **MCS–224**

MASTER OF COMPUTER APPLICATIONS

(MCA–NEW)

Term-End Examination

December, 2024

MCS–224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

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1. (a) Discuss Supervised, Unsupervised and Reinforcement Learning. 5

(b) Compare Artificial Intelligence, Machine Learning and Deep Learning. 5

(c) What is logistic regression ? Briefly discuss the various types of logistic regressions. 5

(d) Let P(x) and Q(x) represent 'x is a rational number' and 'x is a real number', respectively. Symbolize the following sentences : 5

(i) Every rational number is a real number.

(ii) Some real numbers are rational numbers.

(iii) Not every real number is a rational number.

(e) For the following fuzzy set : 5

$$A = \left\{\frac{a}{.5}, \frac{b}{.6}, \frac{c}{.3}, \frac{d}{0}, \frac{e}{.9}\right\}$$

and $$B = \left\{\frac{a}{.4}, \frac{b}{.5}, \frac{c}{.8}, \frac{d}{.2}, \frac{e}{.9}\right\}$$

find fuzzy sets :

(i) $A \cup B$

(ii) $A \cap B$

(iii) $(A \cap B)'$

(f) State the basic steps of machine learning cycle and briefly discuss them. 5

(g) Discuss ensemble-learning. 5

(h) Discuss the advantages and disadvantages of dimensionality reduction. 5

2. (a) Explain K-means clustering algorithm with the help of a suitable example. 10

(b) Explain Single linkage, Complete linkage and Average linkage with a suitable example. 10

3. (a) For the following given transaction data-set, generate association rules using Apriori algorithm. Assume Support = 20%, Confidence = 40% : 10

Transaction-Id	**Set of Items**
T_1	A, B, C
T_2	B, D
T_3	B, E
T_4	A, B, D
T_5	A, E
T_6	B, E
T_7	A, E
T_8	A, B, C, E
T_9	A, B, E

(b) Consider the two-dimensional patterns (2, 1), (3, 5), (4, 3), (5, 6), (6, 7), (8, 8), (9, 10) and (7, 8). Using PCA algorithm, calculate the principal component. 10

4. (a) Apply KNN classification algorithm to the following training data to classify class of X = (10, 7). (Assume K = 3 and use Euclidean distance measure) : 10

x	*y*	**Class**
1	1	A
2	3	A
2	4	A
5	3	A
8	6	B
8	8	B
9	6	B
11	7	B
2	2	A
10	6	B

(b) With the help of suitable example, prove the following properties of fuzzy sets : 10

(i) Commutative

(ii) Associative

(iii) Distributive

(iv) De-Morgan's laws

5. (a) Draw a semantic network for the following English statement : 10

"Mohan greets Nita and Nita's mother greets Mohan."

(b) What are the various factors used to be taken into consideration while developing a state space representation ? 10

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MASTER OF COMPUTER APPLICATIONS [MCA (NEW)]

Term-End Examination

June, 2025

MCS-224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Weightagte : 70%

***Note** : Question No. **1** is compulsory. Attempt any **three** questions from the rest.*

---

1. (a) Compare Artificial Narrow Intelligence, Artificial General Intelligence and Artificial Super Intelligence. 5

   (b) Discuss descriptive, predictive and prescriptive analytics in machine learning. 5

   (c) Differentiate between linear regression and polynomial regression techniques. 5

(d) Let C(x) mean "x is a used car dealer", and H(x) mean "x is honest". Translate each of the following into English sentences : 5

(i) $(\forall x)C(x) \rightarrow \sim H(x)$

(ii) $(\exists x)(C(x) \wedge H(x))$

(iii) $(\exists x)(H(x) \rightarrow C(x))$

(e) With reference to D-S (Dempster Shafer) theory, what are the different types of evidences ? Explain them briefly. 5

(f) Briefly discuss the concept of classification, regression and clustering. Also give the list of algorithms for each concept. 5

(g) What is the purpose of feature extraction in machine learning ? Explain in brief. 5

(h) What is the advantage of using Princer algorithm over Apriori algorithm ? Explain in brief. 5

2. (a) Differentiate between Hierarchical clustering and Partition-based clustering with a suitable example of each concept. 10

(b) Let the following points are to be clustered into 3 groups : 10

$A_1(2,11)$, $A_2(2,15)$, $A_3(8,5)$, $A_4(6,8)$, $A_5(7,9)$, $A_6(6,3)$, $A_7(1,4)$, $A_8(4,8)$

Assume that the initial cluster centers are $A_1(2,11)$, $A_3(8,5)$ and $A_8(4,8)$. Using the Manhattan distance measure and K-mean clustering algorithm, calculate the cluster heads after third iteration.

3. (a) For the following given transactions Data_set, generate association rules

using Apriori algorithm. Assume support = 50% and confidence = 75% :10

Transaction_Id	Set of Items
T_1	Pen, Notebook, Pencil, Colours
T_2	Pen, Notebook, Colours
T_3	Pen, Eraser, Scale
T_4	Pen, Colours, Eraser
T_5	Notebook, Colours, Eraser

(b) Let the training dataset for tissue paper whether it is 'Poor' or 'Fine' based on two properties (acid durability and strength) are as follows : 10

Acid Durability (Sec.)	Strength (gm/cm^2)	Class
7	7	Poor
7	4	Poor
3	4	Fine
2	4	Fine

Suppose there is an unknown sample $X = (3, 7)$, where 3 is acid durability and 7 is strength, find its class using KNN algorithm. (Assume K = 3 and use Euclidean distance).

4. (a) Write ID3-algorithm for creating decision tree for any training dataset. 10

(b) Explain Dempster Shafer theory with a suitable example. 10

5. (a) Suppose a six-faced die is thrown twice. Describe each of the following events : 10

(i) The maximum score is 6.

(ii) The total score is 9.

(iii) Each throw results in an even score larger than 2.

(b) Consider the following missionaries and cannibal problem : 10

Three missionaries and three cannibals are side of a river, along with a boat that can hold one or two people. Find a way to get everyone to the other side, without ever leaving a group of missionaries out-numbered by cannibals.

(i) Formulate and solve the above problem.

(ii) Draw the state-space search graph for solving this problem.

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MCS-224

MASTER OF COMPUTER APPLICATIONS (MCA-NEW)

Term-End Examination

June, 2022

MCS-224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *Question No. 1 is compulsory. Attempt any* ***three*** *questions from the remaining questions.*

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1. (a) Compare Narrow AI, General AI and Super AI. Give suitable example for each. 6

   (b) Briefly discuss the formulation of state space search, with a suitable example. 5

P. T. O.

(c) Write DFS (Depth-First Search) algorithm. Give ***one*** advantage and ***one*** disadvantage of DFS. 5

(d) Obtain Disjunctive Normal Form for the well form formula given below : 5

$$\sim (A \rightarrow (\sim B \wedge C))$$

(e) Briefly discuss the relevance of resolution and unification mechanism in artificial intelligence. 4

(f) Compare supervised learning and unsupervised learning. List the algorithms for each type of learning. 5

(g) Write Bayes' theorem and elaborate the meaning of each component in it. 5

(h) What is Dimensionality Reduction ? Briefly discuss the ways to achieve it. 5

2\. (a) What do you understand by the term "Agents" in Artificial Intelligence ? List the properties, supposed to be possessed by agents. Also, compare the SR (simple reflex) agents with model based reflex agents. 10

(b) Draw the block diagram of machine learning cycle and briefly discuss the role of each of its components. 10

3. (a) What is Adversarial Search ? How it is different from the normal search ? Briefly discuss the types of adversarial search. 10

(b) Differentiate between the following, with an example for each : 10

(i) Machine learning and Data mining

(ii) Multi-class classification and Multi-label classification

4. (a) What is Best First Search ? How is it different from greedy best first search ? Give time complexity and space complexity of best first search. Also, give advantage and disadvantage of best first search. 10

(b) How regression differs from classification ? What is the similarity between the two ? Briefly discuss the following type of regression : 10

(i) Linear regression

(ii) Multiple linear regression

(iii) Logistic regression

(iv) Polynomial regression

5. Write short notes on any ***five*** of the following :

5×4=20

(a) Rule based systems and its types

(b) Frames

(c) Dempster Scheffer theory

(d) Closed world assumption

(e) Recursive neural networks

(f) Generative adversarial networks

(g) Support vector machines

(h) Confusion matrix

(i) Linear discriminant analysis

(j) Pincer search

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MCS-224

MASTER OF COMPUTER APPLICATIONS (MCA) (NEW)

Term-End Examination

December, 2022

MCS-224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *Question **No. 1** is compulsory. Attempt any **three** questions from the rest.*

---

1. (a) Compare Artificial Intelligence (AI), Machine Learning and Deep Learning. 6

   (b) Briefly discuss the Adversarial search. Name the techniques used for adversarial search. 5

   (c) Write algorithm for BFS (Breadth-First Search). Write the time complexity and space complexity of BFS. 5

(d) Obtain Conjunctive Normal Form (CNF) for the formula : 5

$$D \rightarrow (A \rightarrow (B \wedge C))$$

(e) What is Skolomization ? Skolomize the expression : 4

$$(\exists_{X_1})(\exists_{X_2})(\forall_{Y_1})(\forall_{Y_2})(\exists_{X_3})(\forall_{Y_3})$$

$$P(X_1, X_2, X_3, Y_1, Y_2, Y_3)$$

(f) What is Reinforcement Learning ? Classify the various reinforcement learning algorithms. 5

(g) What is Logistic Regression ? Briefly discuss the various types of logistic regressions. 5

(h) Differentiate between linear regression and polynomial regression techniques. 5

2. (a) In context of Intelligent Agents, what are task environments ? Explain the standard set of measures for specifying a task environment under the heading PEAS. 10

(b) Briefly discuss the following (give suitable example for each) : 10

(i) Rote learning

(ii) Supervised learning

(iii) Unsupervised learning

(iv) Delayed-Reinforcement learning

3. (a) Briefly discuss the Min-Max Search Strategy. What are the properties of Minimax Algorithm ? Also give advantages and disadvantages of Minimax search. 10

(b) Differentiate between the following, with an example for each : 10

(i) Classification techniques and Regression techniques

(ii) Lazy learner algorithms and Eager learner algorithms

4. (a) What is Iterative Deepening Depth First Search (IDDFS) ? How is it different from Depth First Search ? Give time and space complexities of IDDFS. Also give advantages and disadvantages of IDDFS. 10

(b) Discuss support vector regression. Draw suitable diagram in support of your discussion. Also give ***two*** applications of support vector regression. 10

5. Write short notes on any ***five*** of the following : 5×4=20

(a) Forward Chaining

(b) Semantic Nets

(c) Bayes' Networks

(d) Rough Set Theory

(e) Recurrent Neural Networks

(f) Restricted Boltzmann Machines

(g) Ensemble Methods

(h) K-Nearest Neighbour

(i) Principal Component Analysis

(j) Association Rules

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MASTER OF COMPUTER APPLICATIONS (MCA-NEW)

Term-End Examination

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MCS-224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Weightage : 70%

Note : *Question No. **1** is compulsory. Attempt any **three** questions from the remaining questions.*

---

1. (a) Compare descriptive, predictive and prescriptive analytics in machine learning. 6

   (b) What is Min-Max Search Strategy ? Write MINIMAX algorithm. 5

(c) Differentiate between informed search and uninformed search. Name *one* algorithm for each. 4

(d) Describe the Modus Ponens and Modus Tollens as propositional rule of inference. 5

(e) What is Prenex Normal Form (PNF) ? Transform the following formula into PNF : 5

$$(\forall_x)(Q(x) \rightarrow (\exists_x)R(x, y))$$

(f) What is Ensemble Learning ? Briefly discuss any *one* of the ensemble learning method. 5

(g) Draw confusion matrix and write formula for accuracy, precision, sensitivity and specificity. 5

(h) What is a Neural Network ? How biological neuron relates to Artificial Neuron ? Illustrate with suitable diagram and a table to map the components of Biological Neuron with Artificial Neuron. 5

2. (a) Explain Turing test, with the help of a block diagram. Also, discuss Chinese room test as criticism to Turing test. 10

(b) Briefly discuss the following, with suitable example for each : 10

(i) Rule-based machine learning

(ii) Bayesian Algorithms

(iii) Decision trees

(iv) Dimensionality reduction

3. (a) What do you understand by State Space Search ? Explain the state space representation of Water-Jug Problem (WJP), given below :

"Given two jugs of 5-gallon and 3-gallon, both of which do not have measuring indicators on them. The jugs can be filled with water with the help of any pump, any number of times."

The question is "how can you get 4 gallons of water in a 5-gallon jug ?" 10

(b) What is Binary Classification ? Can binary classification algorithms be altered to work for problems with more than two classes ? Justify. Also, discuss one-versus the rest and one-versus-one approach. 10

4. (a) Differentiate between the following : 10

(i) A* and AO* algorithm

(ii) Depth first search and Breadth first search

(b) What is linear regression ? How linear regression is performed using least square method ? Find the regression line for the data points (x, y) tabulated below : 10

x	y
1	3
2	4
3	2
4	4
5	5

Also, discuss the terms 'mean squared error' and 'mean absolute errors'.

5. Write short notes on any ***five*** of the following : $5 \times 4 = 20$

(a) Backward Chaining

(b) Scripts

(c) Non-Monotonic Reasoning Systems

(d) Convolutional Neural Networks

(e) Auto Encoders

(f) Transformers

(g) Logistic Regression

(h) Naive Bayes Algorithm

(i) Feature Selection

(j) K-Means Algorithm

No. of Printed Pages : 4 **MCS–224**

MASTER OF COMPUTER APPLICATIONS (MCA-NEW)

Term-End Examination

December, 2023

MCS-224 : ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 Hours *Maximum Marks : 100*

Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

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1. (a) Differentiate between clustering and classification technique. List any *two* algorithms for each. 6

   (b) Explain Turing test with suitable example. 5

(c) What is skolomization ? What is the utility of skolomization ? Skolomize the following expression : 6

$$\exists_{x_1}\exists_{x_2}\forall_{y_1}\forall_{y_2}\exists x_3\forall_{y_3}P(x_1, x_2, x_3, y_1, y_2, y_3)$$

(d) Explain backward chaining system with a suitable example. 6

(e) Differentiate between Lazy learners and Eager learners in classification problem. Also, list the algorithms used for each type of learners, respectively. 6

(f) Explain the term 'Dimensionality Reduction'. Name the general techniques used to perform it. Also give merits and limitation of dimensionality reduction. 6

(g) Explain the working of partition based clustering. Mention any *two* methods used for partition based clustering. 5

2. (a) Compare artificial intelligence, machine learning and deep learning. 6

(b) What do you understand by state space in AI ? What is its utility ? Write production rules for state space representation of water jug problem. 7

(c) Write and explain Breadth First Search (BFS) algorithm. Discuss its space and time complexity. Also, give advantage and disadvantage of BFS algorithm. 7

3. (a) Differentiate between predicate and propositional logic. If $P(x) \rightarrow$ "x is a rational number" and $Q(x) \rightarrow$ "x is a real

number" then symbolize the following sentences : 5

(i) Every rational number is a real number

(ii) Some real numbers are rational

(iii) Not every real number is a rational number

(b) Explain the concept of resolution and unification in AI, with suitable example for each. 7

(c) Explain rule based systems in AI. Give advantages and disadvantages of rule based systems. Also, give the *two* important sources of uncertainty in rule based systems. 8

4. (a) Explain reinforcement learning with the help of a block diagram. Explain the role of each component of block diagram. 6

(b) Compare classification and regression techniques of supervised learning. Explain the various metrics used for evaluating the classification model. 7

(c) Differentiate between the following (give example for each) : 7

(i) Logistic regression and Linear regression

(ii) K-NN algorithm and K-means algorithm

5. Explain any ***four*** of the following with suitable example for each : 4×5=20

(a) Linear Discriminant Analysis

(b) FP tree growth

(c) Density based clustering

(d) Restricted Boltzmann Machines

(e) Convolutional Neural Networks

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Time : 3 Hours *Maximum Marks : 100*

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Note : *Question No.* ***1*** *is compulsory. Attempt any* ***three*** *questions from the rest.*

1. (a) Compare Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI) and Artificial Super Intelligence (ASI). 6

P. T. O.

(b) Explain Chinese room test as Criticism to Turing test, with suitable example. 5

(c) Write steps to transform FOPL (First Order Predicate Logic) to PNF (Prenex Normal Form). Apply the steps to transform $\forall_x (Q(x) \rightarrow \exists_x R(x, y))$ to PNF. 6

(d) Explain forward chaining systems with suitable example. 6

(e) Draw block diagram for the machine learning cycle. Also list the steps involved in machine learning cycle. 6

(f) Explain the working of FP growth algorithm. Give advantages of FP growth over Apriori Algorithm. 6

(g) Compare clustering and classification. Give suitable example for each. Also, list the algorithms for each. 5

2. (a) Compare descriptive, predictive and prescriptive analytics, performed under machine learning. 6

(b) Explain Min-Max search algorithm with suitable example. Give properties of Min-Max search algorithm. Also, write its advantages and disadvantages. 7

(c) Write and explain Depth First Search (DFS) algorithm. Give time and space complexity of DFS algorithm. Also give its advantage and disadvantage. 7

3. (a) Given C(x) = "X is a used car dealer", and H(x) = "X is honest" then translate the following into English sentences : 5

(i) $\exists_x$ C(x)

(ii) $\exists_x$ H(x)

(iii) $\forall_x$ C(x) $\rightarrow \sim$ H(x)

(iv) $\exists_x$ (C(x) $\wedge$ H(x))

(v) $\exists_x$ (H(x) $\rightarrow$ C(x))

(b) What do you understand by the term 'Resolution' in AI ? Discuss the utility of Resolution mechanism in AI. Apply it to conclude 'Raman is mortal' from the knowledge given below : 7

(i) Every man is mortal

(ii) Raman is a man

(c) Write short notes on any *two* of the following, with suitable example for each : 8

(i) Frames

(ii) Scripts

(iii) Semantic nets

4. (a) Compare model-free reinforcement learning with mode-based reinforcement learning. Also, discuss the sub-classes of model-free reinforcement learning. 6

(b) Write and explain Bayes' theorem. Also, write and explain Naive Bayes' algorithm with suitable example. 7

(c) Write short notes on any *two* of the following (give example for each) : 7

(i) Support Vector Machines

(ii) Support Vector Regression

(iii) Polynomial Regression

5. Explain any ***four*** the following with suitable example for each : 4×5=20

(a) Principal Component Analysis

(b) Apriori Algorithm

(c) Hierarchical Clustering

(d) Generative Adversarial Networks

(e) Auto Encoders